

Wind speed (m/s)	Blade angle ($^{\circ}$)	Rotor speed (RPM)	Power Output (kW)
5	11	125	48
7	9	155	72
9	15	175	101
11	13	220	142
13	18	235	176

$$y = m_1 x_1 + m_2 x_2 + m_3 x_3 + C$$

$$\theta = (X^T X)^{-1} X^T y$$

$$X = \begin{bmatrix} 1 & 5 & 11 & 125 \\ 1 & 7 & 155 \\ 1 & 9 & 175 \\ 1 & 11 & 220 \\ 1 & 13 & 235 \end{bmatrix} \quad y = \begin{bmatrix} 48 \\ 72 \\ 101 \\ 142 \\ 176 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 5 & 45 & 66 & 910 \\ 45 & 445 & 630 & 8760 \\ 66 & 630 & 920 & 12485 \\ 910 & 8760 & 12485 & 173900 \end{bmatrix}$$

$$A = \begin{bmatrix} 5 & 45 & 66 & 910 \\ 45 & 445 & 630 & 8760 \\ 66 & 630 & 920 & 12485 \\ 910 & 8760 & 12485 & 173900 \end{bmatrix}$$

$$A^{-1} = \frac{\text{adj } A}{|A|}$$

$$|A| = 196600$$

~~$$A^{-1} = \begin{bmatrix} 76.901704 & 18.214776 & -3.0043 \\ 18.2147 & 4.5863 & -0.7240 \\ -3.0043 & -0.7240 & 0.1602 \\ -1.1042 & -0.2743 & 0.0406 \end{bmatrix}$$~~

$$A^{-1} = \begin{bmatrix} 76.9017 & 18.2147 & -3.0043 & -1.1042 \\ 18.2147 & 4.5863 & -0.7240 & -0.2743 \\ -3.0043 & -0.7240 & 0.1602 & 0.0406 \\ -1.1042 & -0.2743 & 0.0406 & 0.0166 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 1 & 1 & 6 & 1 & 7 & 1 \\ 5 & 7 & 9 & 11 & 13 \\ 11 & 9 & 15 & 13 & 18 \\ 125 & 155 & 175 & 220 & 235 \end{bmatrix} \begin{bmatrix} 48 \\ 72 \\ 101 \\ 142 \\ 176 \end{bmatrix} = Y^T X$$

$$= \begin{bmatrix} 539 \\ 5503 \\ 7705 \\ 107435 \end{bmatrix}$$

$$\Theta = \begin{bmatrix} 76.901704 & 18.214776 & -3.004323 & -1.104273 \\ 18.214776 & 4.586343 & -0.724059 & -0.274364 \\ -3.004323 & -0.724059 & 0.160224 & 0.040692 \\ -1.104273 & -0.274364 & 0.040692 & 0.016684 \end{bmatrix} \begin{bmatrix} 539 \\ 5503 \\ 7705 \\ 107435 \end{bmatrix}$$

$$= \begin{bmatrix} -9.811236 \\ -2.214906 \\ 1.166605 \\ 0.484357 \end{bmatrix}$$

$$\therefore y = \cancel{-9.811236 x_1} + \cancel{(-2.214906) x_2} + \cancel{1.166605 x_3}$$

+

$$\therefore y = (-2.214906 x_1) + (1.166605 x_2) + (0.484357 x_3) - 9.811236$$

Let $x_1 = 9$, $x_2 = 11$ and $x_3 = 100$

Then,

$$\begin{aligned} y &= (-2.214906 \times 9) + (1.166605 \times 11) + (0.484357 \times 100) \\ &\quad - 9.811236 \\ &= \underline{\underline{31.530579}} \end{aligned}$$